

Hands-on Lab: Building a Joule Agent with SAP Build

Aleksandrs Antonuks, Vass, aleksandrs.antonuks@vasscompany.com

Michael Pytel, Vass, michael.pytel@vasscompany.com

Sheldon Lipshitz, Vass, sheldon.lipshitz@vasscompany.com

<https://vasscompany.com>

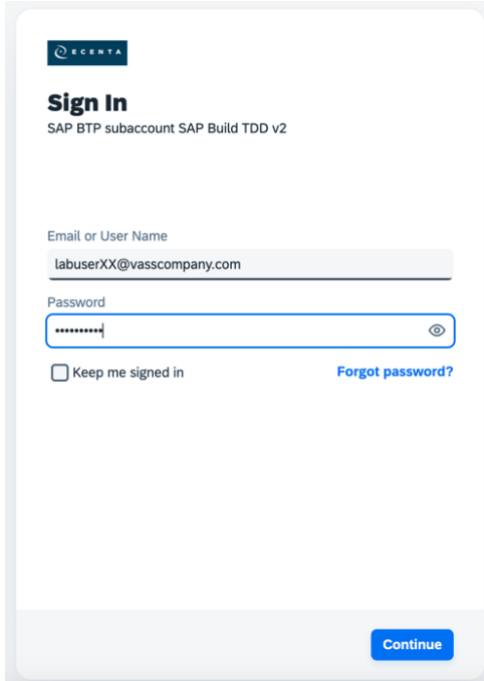
Index

Table of Contents

Environment Information	2
Lab Goal	3
Create a Joule Skills	3
Create a Joule Agent	16
Test your Joule Agent	18

Environment Information

1. Please use GOOGLE CHROME as your browser
2. SAP Build App URLs
 - o [SAP Build Apps](#)
3. Lab User Information
 - o User: **labuser<##>@vasscompany.com**
 - o Password: Newpa55!
4. Replace <##> with your Lab User Number assigned by the instructors



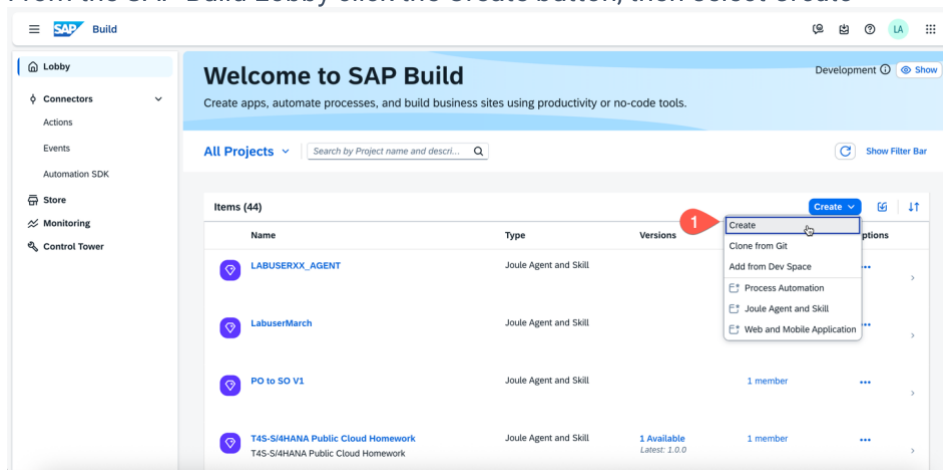
The screenshot shows a 'Sign In' page for 'SAP BTP subaccount SAP Build TDD v2'. The page has a white background with a blue header bar containing the 'SAP' logo. Below the header, the text 'Sign In' is displayed in bold, followed by 'SAP BTP subaccount SAP Build TDD v2'. There are two input fields: 'Email or User Name' and 'Password'. The 'Email or User Name' field contains the text 'labuserXX@vasscompany.com'. The 'Password' field contains a series of dots. Below the password field, there is a checkbox labeled 'Keep me signed in' and a link labeled 'Forgot password?'. At the bottom right, there is a blue button labeled 'Continue'.

Lab Goal

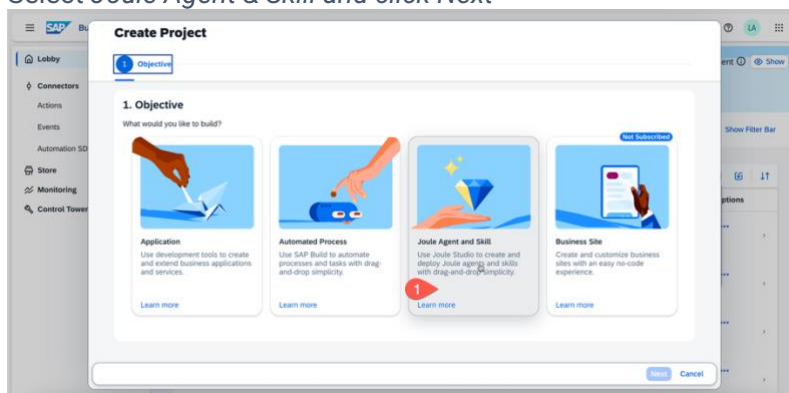
We will be building a Joule Agent that can review Opportunities within Salesforce.com and then create a corresponding Sales Order within S/4HANA. For this lab, we have already set up the 'Actions' which call the API endpoints in Salesforce.com and S/4HANA. Screenshots will be provided in the Lab Overview presentation. This is a very simple, self-contained scenario to demo the low-code environment where developers or analysts can create Joule Agents. This is not a production-ready scenario as all environments are distinctly different and require different inputs. This lab should provide a good, first, hands-on experience to encourage the lab user to explore more.

Create a Joule Skills

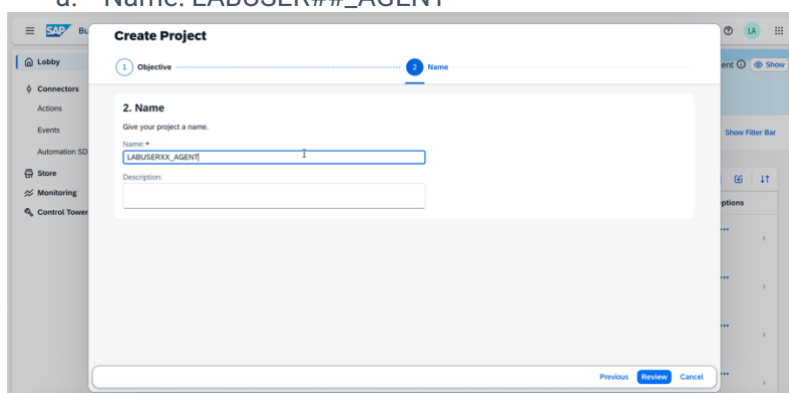
1. From the SAP Build Lobby click the Create button, then select *Create*



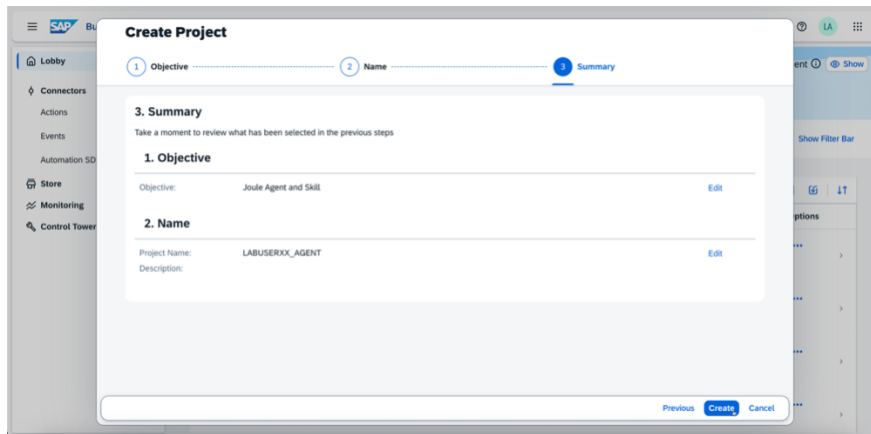
2. Select *Joule Agent & Skill* and click Next



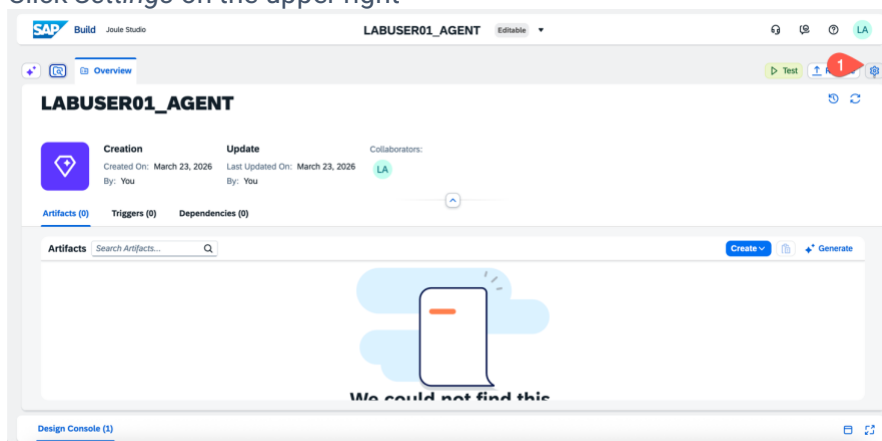
- 4.
5. Enter a name for your Joule Skill in the following format
 - a. Name: LABUSER##_AGENT



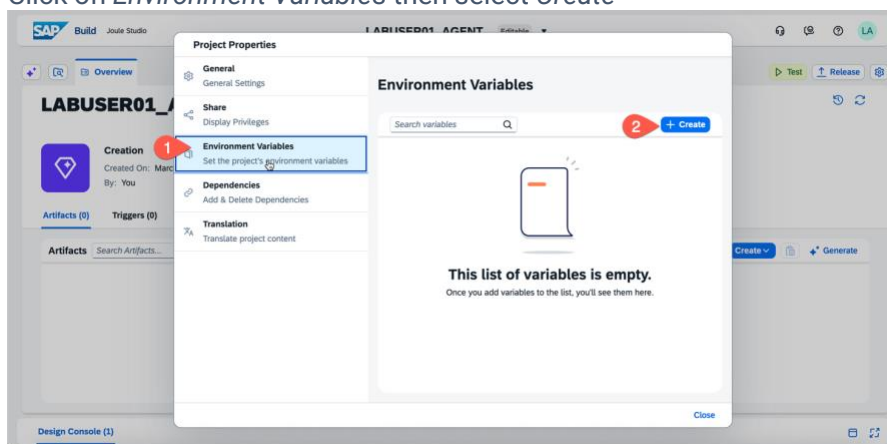
- 6.
7. Click Review. Then click *Create*



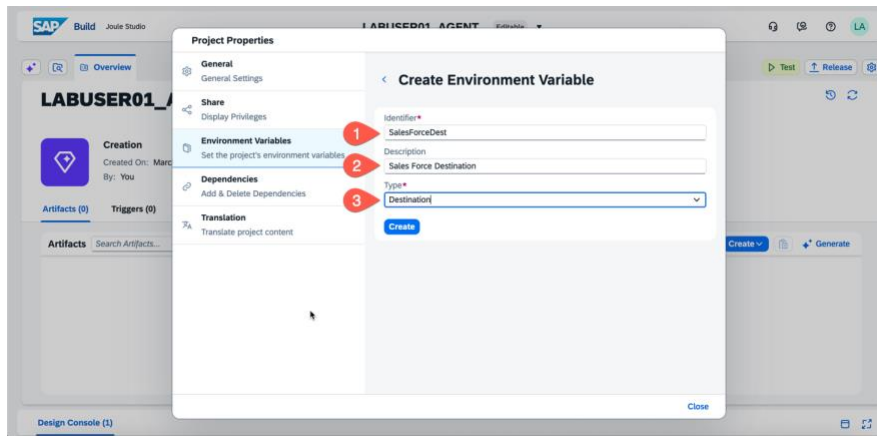
- 8.
9. A new browser tab will open (suppress the pop-up blocker)
10. Accept the Disclaimer if shown.
11. Click **Settings** on the upper right



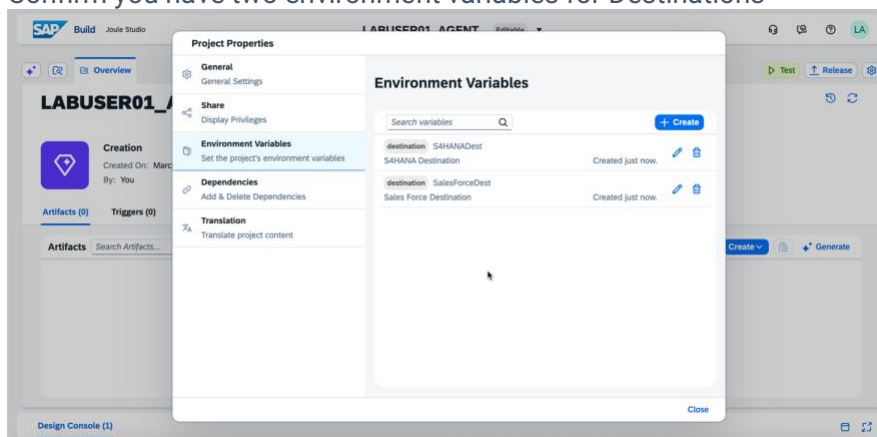
- 12.
13. Click on **Environment Variables** then select **Create**



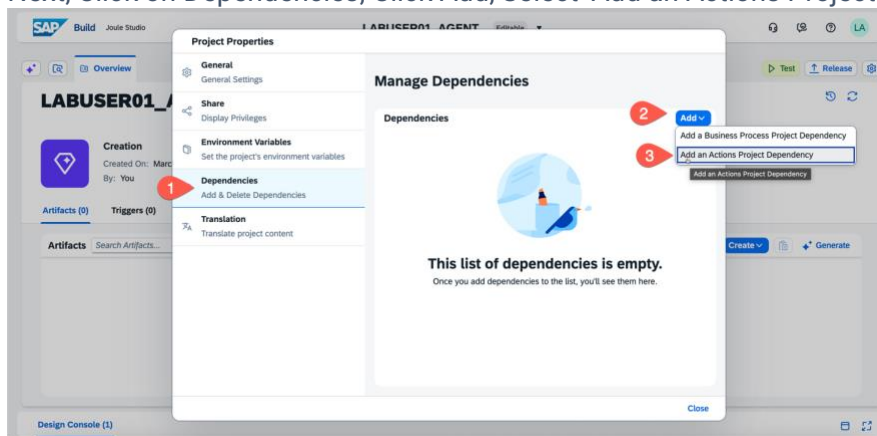
- 14.
15. Create the Destination with the data below and click **Create** again
 - a. Identifier: SalesforceDest
 - b. Description: Sales Force Destination
 - c. Type: Destination



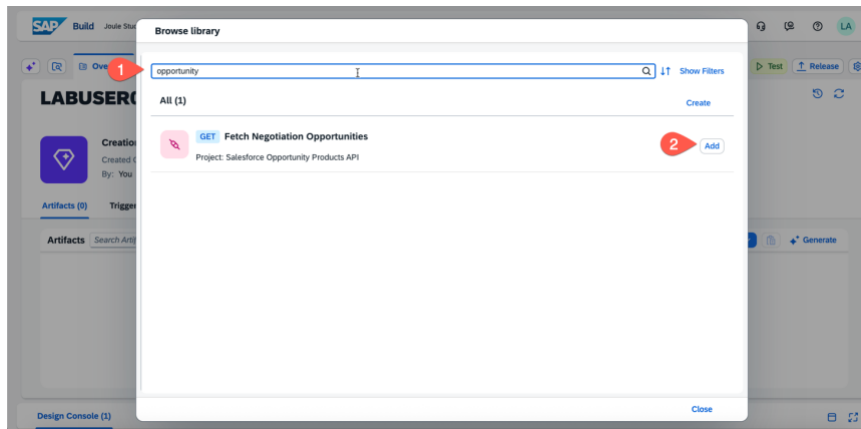
- 16.
17. Repeat for S/4HANA with the data below
 - a. Identifier: S4HANADest
 - b. Description: S4HANA Destination
 - c. Type: Destination
18. Confirm you have two environment variables for Destinations



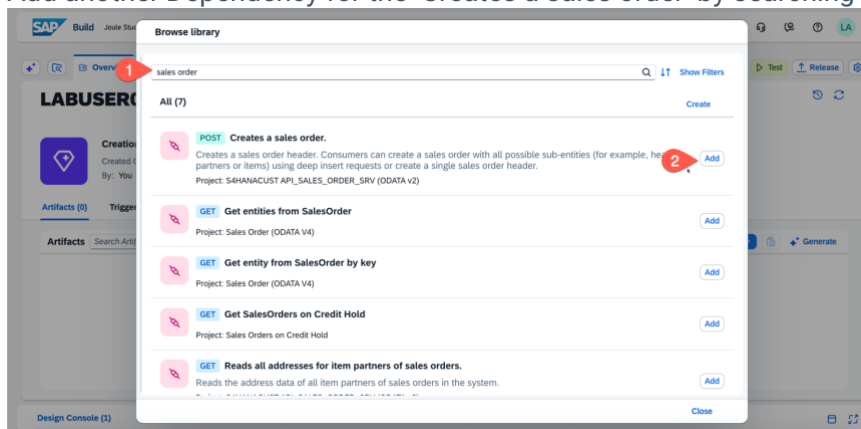
- 19.
20. Next, Click on Dependencies, Click Add, Select 'Add an Actions Project Dependency'



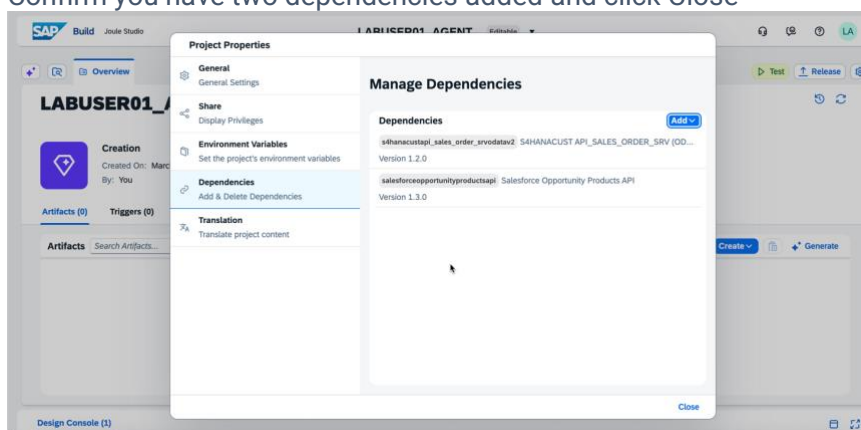
- 21.
22. When prompted to Search, enter 'Opportunity' and select the Add button next to 'Fetch Negotiation Opportunities'



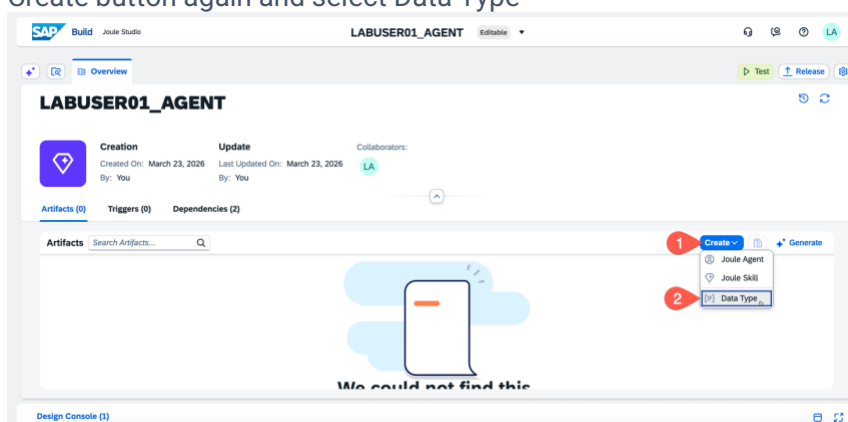
23. Add another Dependency for the 'Creates a sales order' by searching



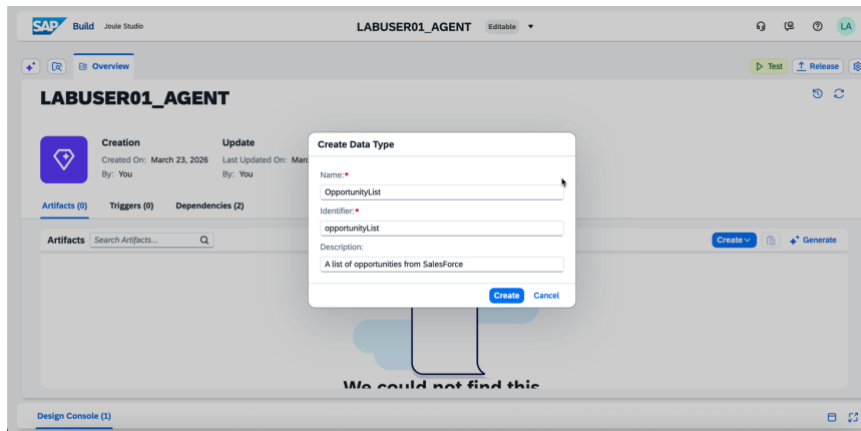
25. Confirm you have two dependencies added and click Close



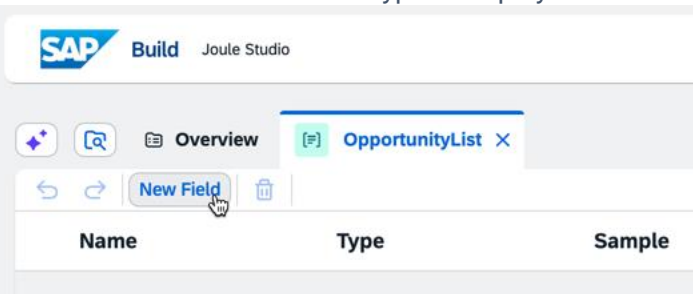
27. Next, we're going to create a Data Type to persist the results of the Salesforce API. Click the Create button again and select Data Type



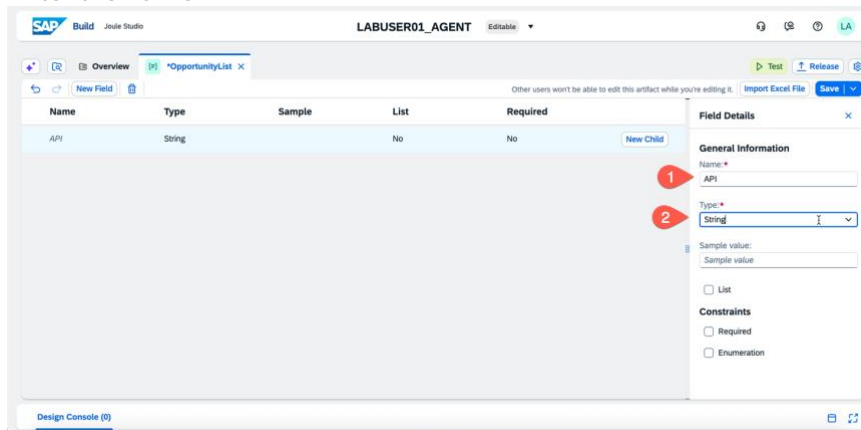
29. Enter the following details and click Create
- Name: OpportunityList
 - Description: A list of opportunities from Salesforce



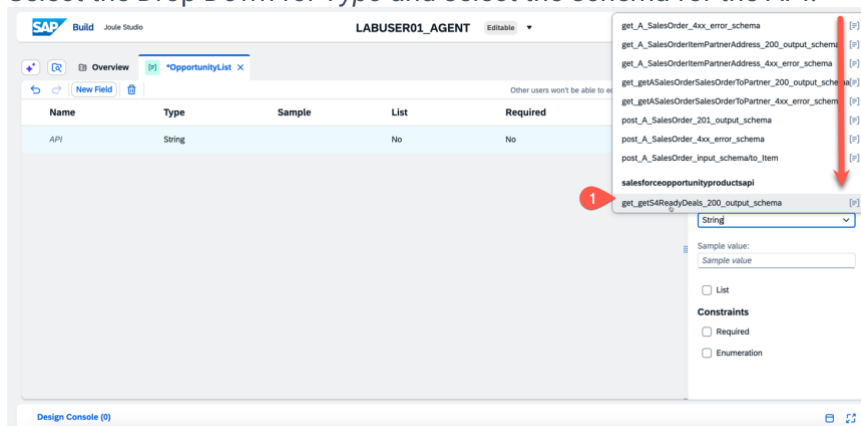
31.
 32. Click New Field when the Data Type is displayed



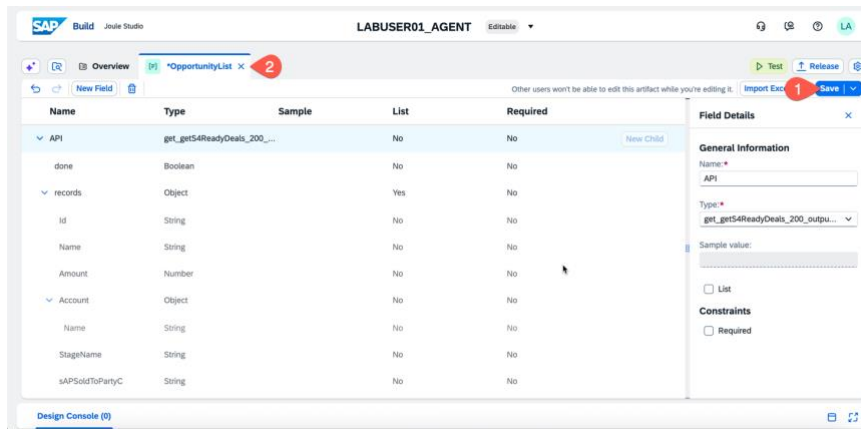
33.
 34. Enter the name 'API'



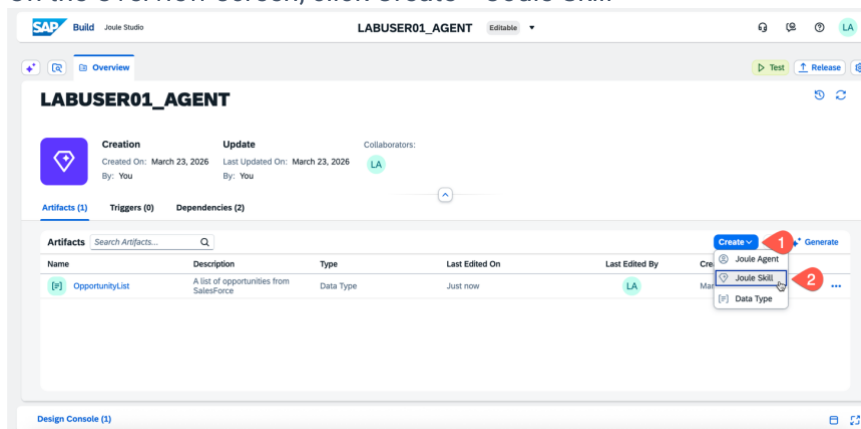
35.
 36. Select the Drop Down for Type and select the schema for the API.



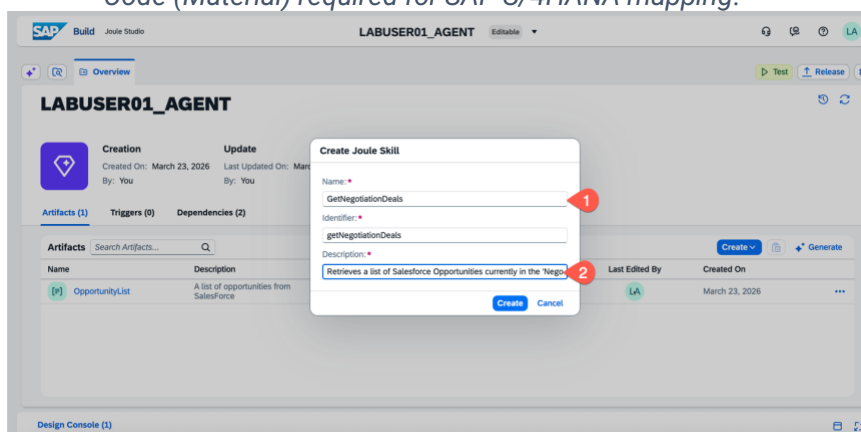
37.
 38. Click Tab Button and the resulting API schema should be displayed.
 39. Click Save and then close the Tab for the Data Type



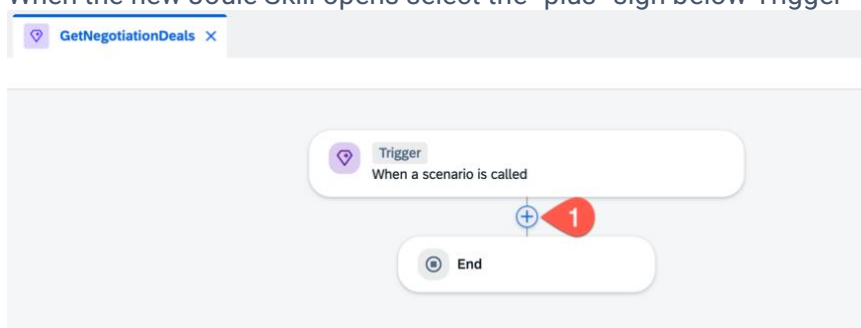
40. On the Overview screen, click Create > Joule Skill



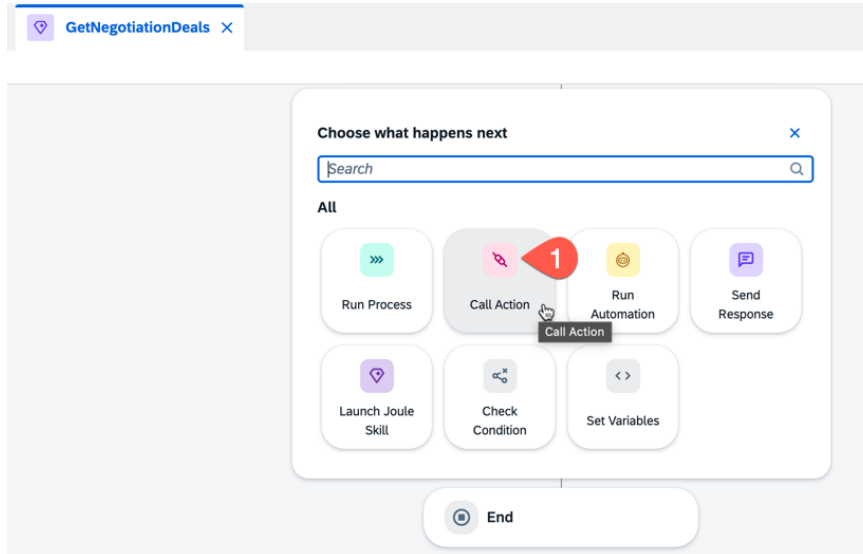
42. Enter the following details on the Joule Skill
- Name: GetNegotiationDeals
 - Description: Retrieves a list of Salesforce Opportunities currently in the 'Negotiation/Review' stage. For each opportunity, it provides the Opportunity ID, the Deal Name, the associated Account Name, and a nested list of products (Line Items). The product data includes the Quantity, Unit Price, Salesforce Product Name, and the Product Code (Material) required for SAP S/4HANA mapping.



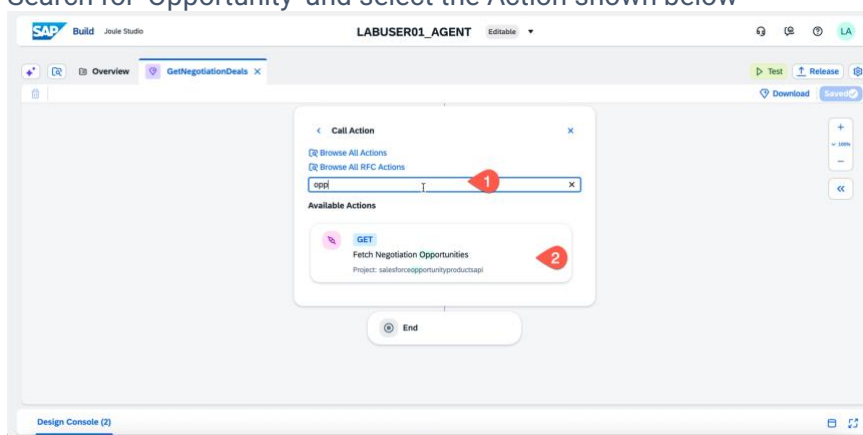
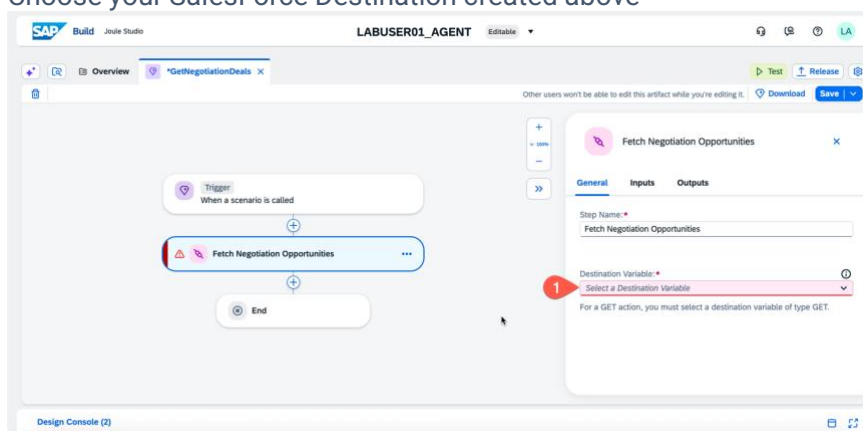
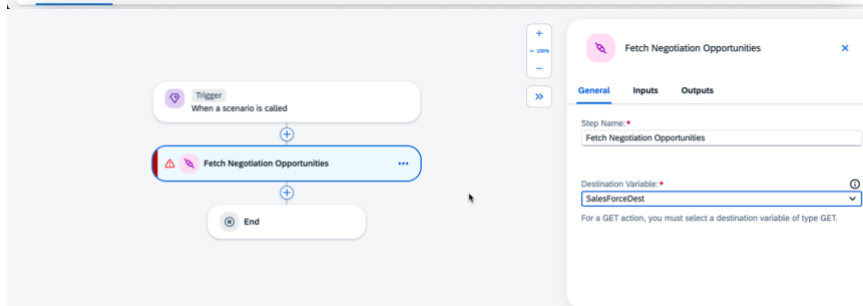
44. When the new Joule Skill opens select the "plus" sign below Trigger

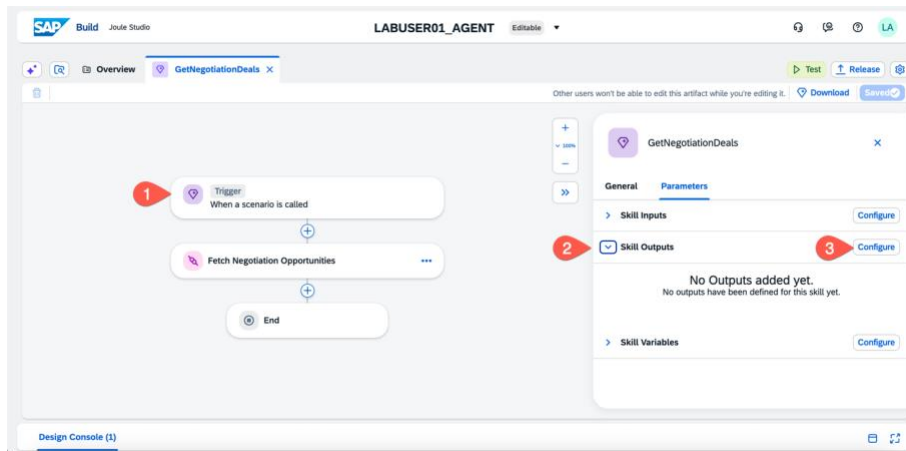


- 46.

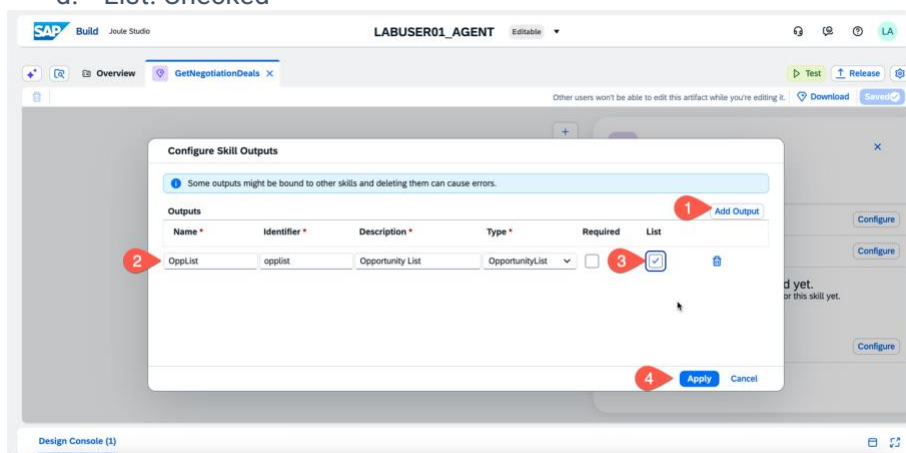
47. Select a *Call Action*

48. Search for 'Opportunity' and select the Action shown below

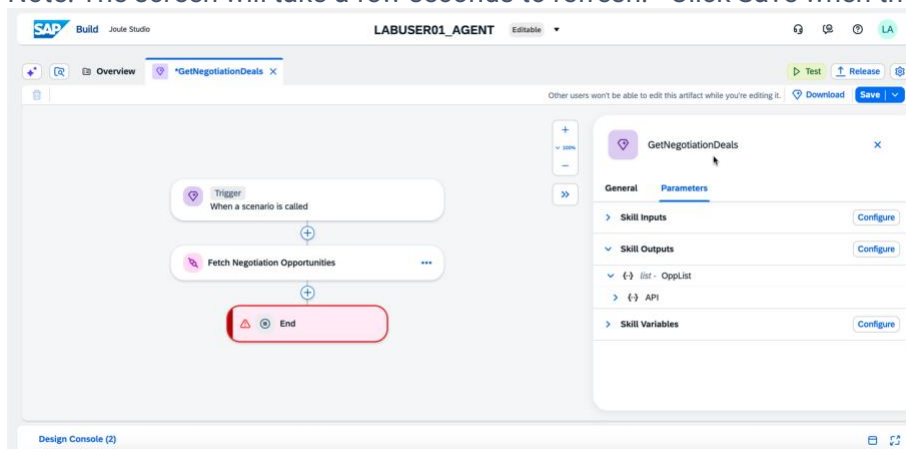

 50. When the Action is added select the Destination dropdown
 51. Choose your SalesForce Destination created above

 53. Click Save in the upper right.
 54. Next, Click on *Trigger* then select Parameters > Skill Outputs > Configure




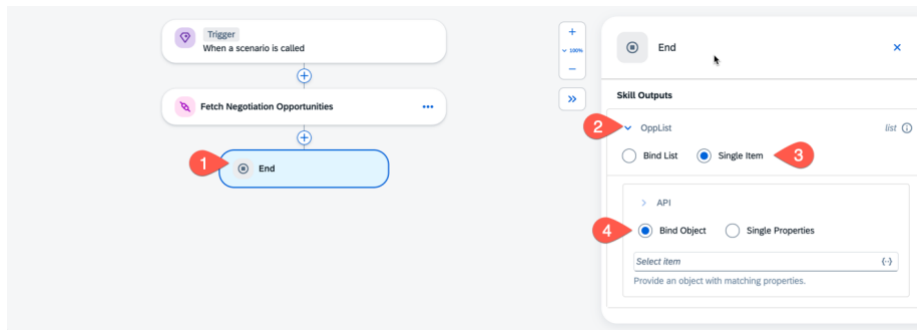
57.
 58. Click *Add Output* and enter the following details
- Name: OppList
 - Description: Opportunity List
 - Type: Scroll to select your Data Type created above
 - List: Checked



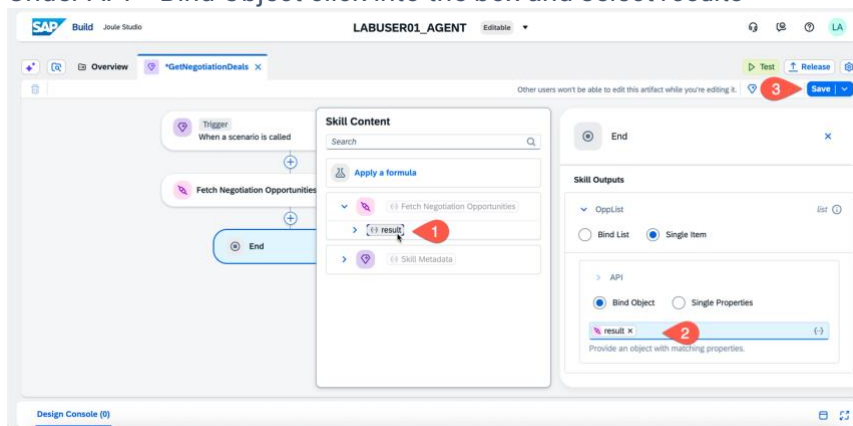
59.
 60. Click *Apply*
 61. Note: The screen will take a few seconds to refresh. Click *Save* when the Output is shown.



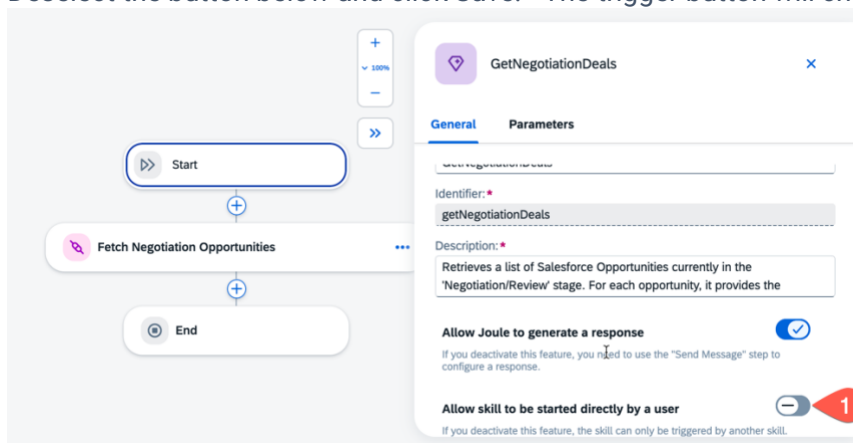
62.
 63. Click on *End* in red
 64. Click on *Opp List*, then select *Single Item*



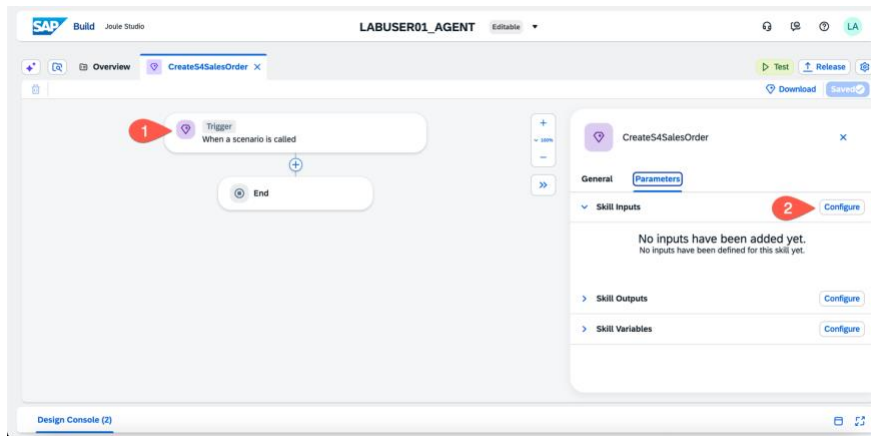
- 65.
66. Under API > Bind Object click into the box and select *results*



- 67.
68. Click Save again
69. Next, we want to click on 'Trigger' and disable this skill from being called directly. This will ensure that Joule does have issues with intent matching.
70. Deselect the button below and click Save. The trigger button will change to a Start Button



- 71.
72. Close the tab for the Joule Skill
73. Click *Create > Joule Skill* again
74. Enter the following details and click Create
- Name: CreateS4SalesOrder
 - Description: *Creates a new Sales Order in SAP S/4HANA based on finalized Opportunity data. It maps the Salesforce 'ProductCode' to the S/4HANA 'Material' field, sets the 'RequestedQuantity', and assigns the 'SoldToParty' based on the SalesForce field SAP_Sold_To_Party__c. This skill should be triggered only after the user confirms the line items retrieved from Salesforce are correct for order entry. The result is a confirmation of the sales order creation with the sales order number from the JSON result.*
75. Select the Trigger, then select Parameters, Skill Inputs and Configure



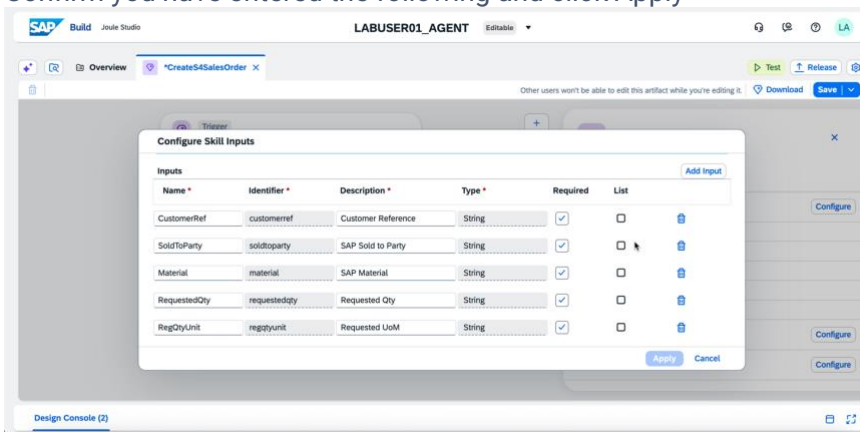
76.

77. Enter the following Input Parameters

Name:	Description
CustomerRef	Customer Reference
SoldToParty	SAP Sold to Party
Material	SAP Material
RequestedQty	Requested Qty
RegQtyUnit	Requested UoM

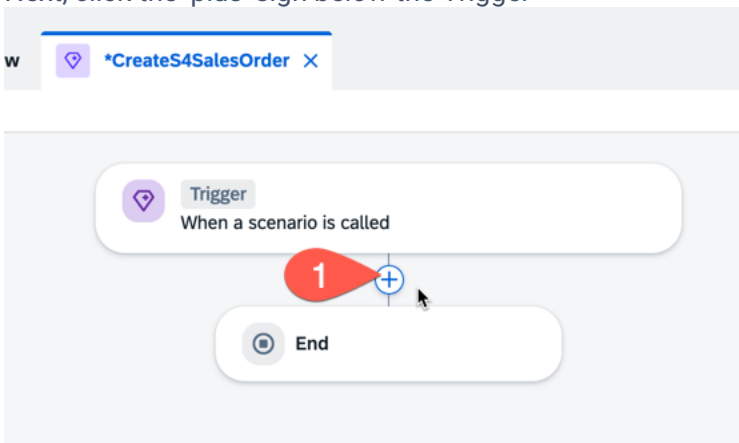
78. Check Required for all parameters

79. Confirm you have entered the following and click Apply



80. Design Console (2)

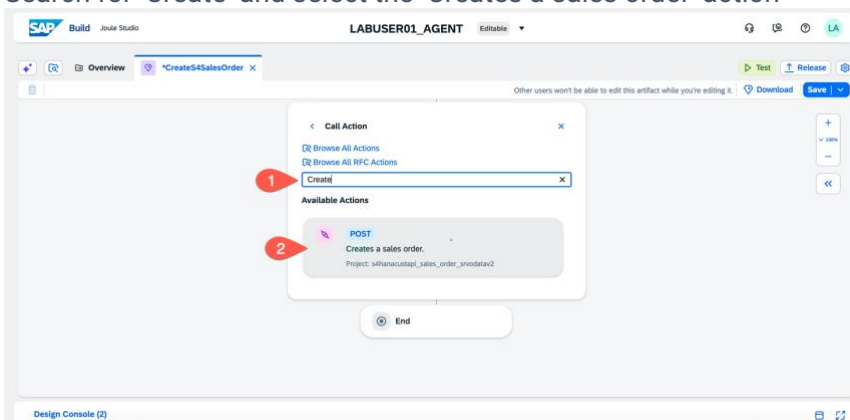
81. Next, click the 'plus' sign below the Trigger



82.

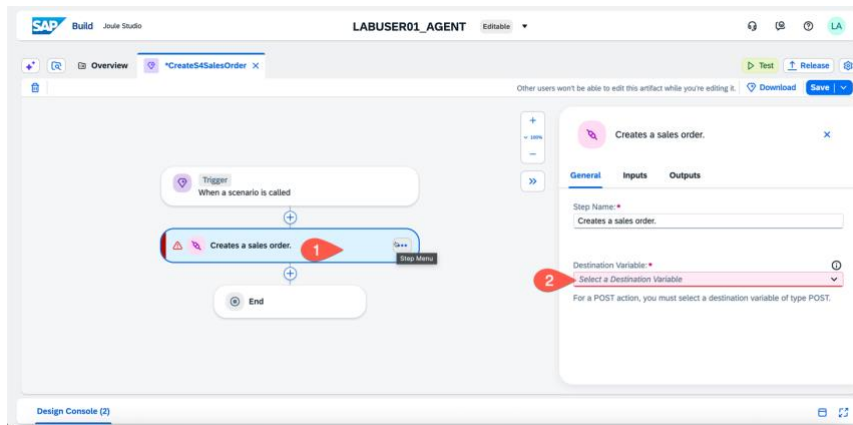
83. Select Call Action

84. Search for 'Create' and select the 'Creates a sales order' action

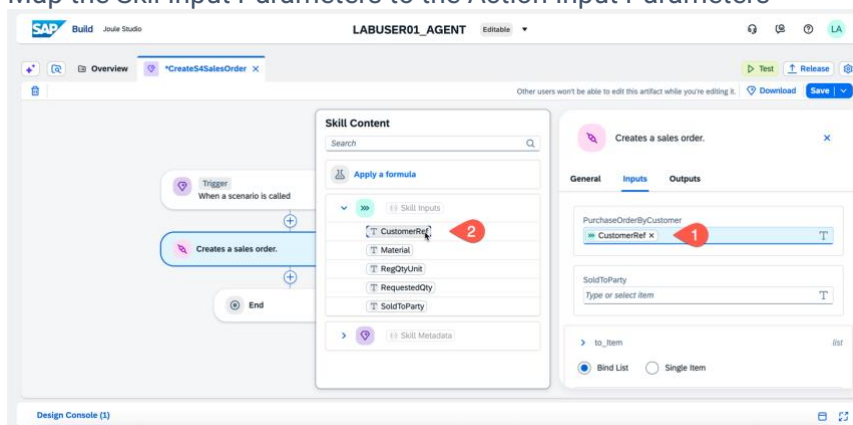


85. Design Console (2)

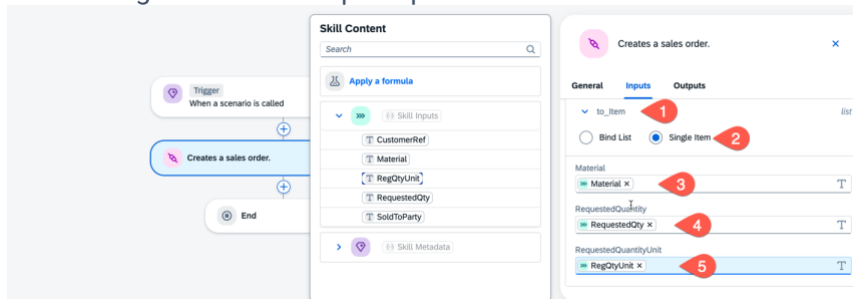
86. Select your S4HANA Destination



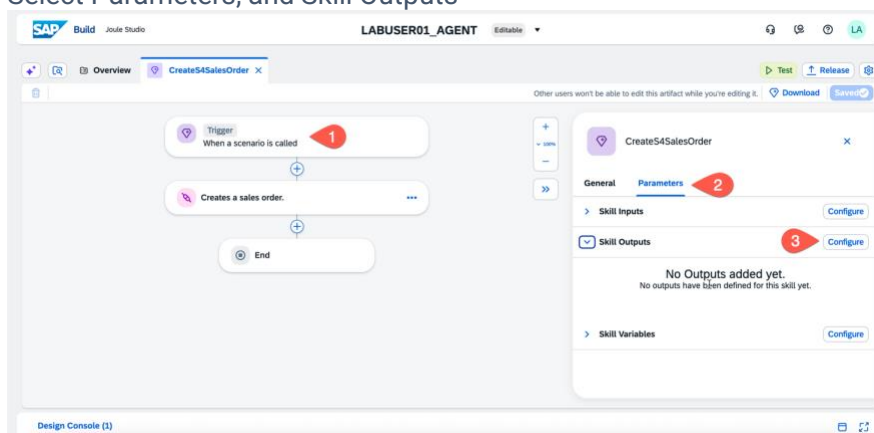
87. Click the Inputs on your Action
88. Map the Skill Input Parameters to the Action Input Parameters



90. Scroll Down to 'to_Item' and open the field
91. Select 'Single Item' and map the parameters

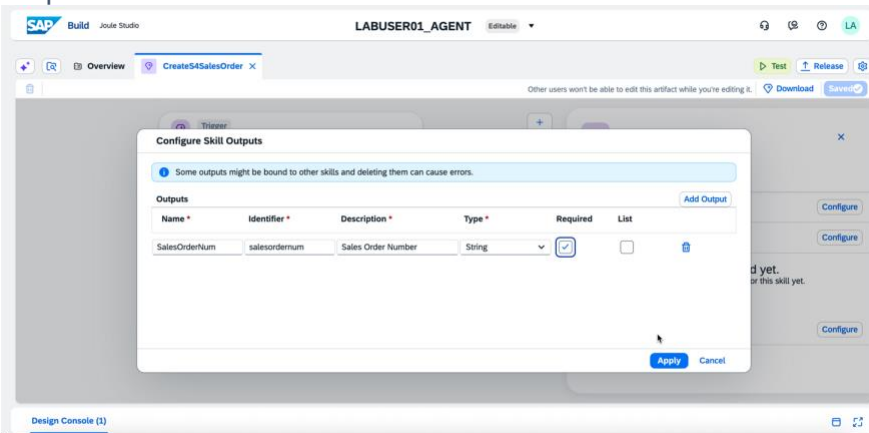


92. Click the Save Button in upper right again
93. After the Save is complete, click on Trigger.
94. Select Parameters, and Skill Outputs



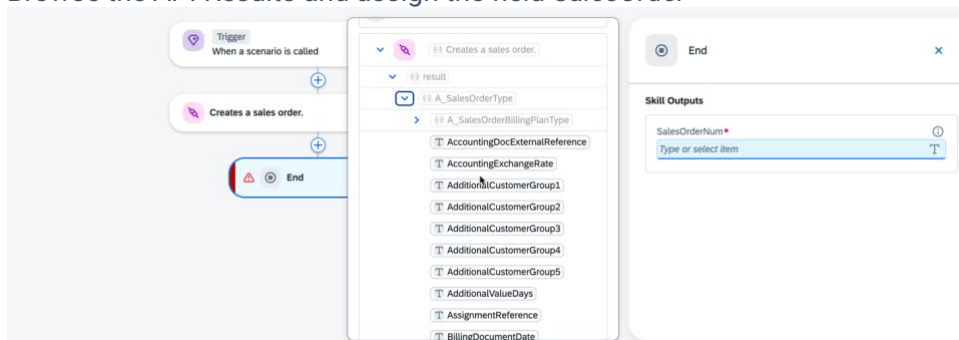
95. Click Configure
96. Click Add Output and enter the following
 - a. Name: SalesOrderNum

- b. Description: Sales Order Number
- c. Type: String
- d. Required: Checked

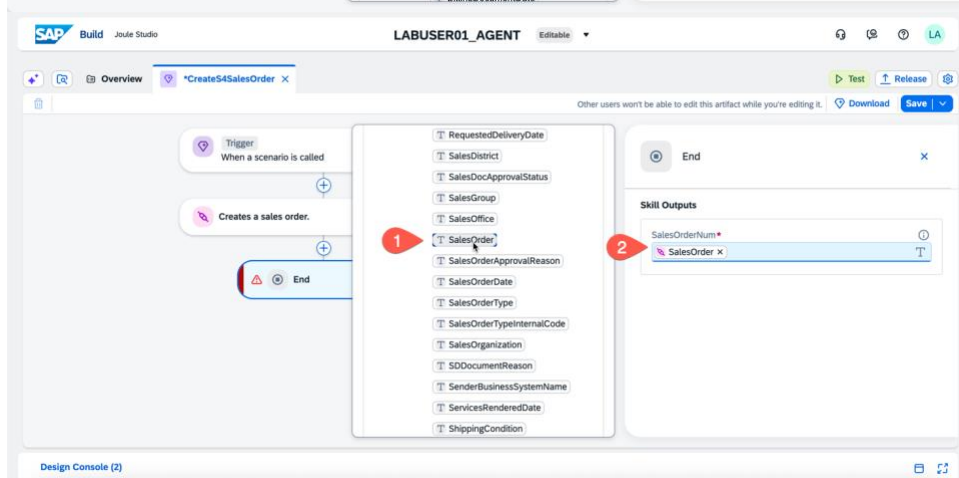


100.
101.
102.
103.

Click Apply
Next, click on End, and click into the field SalesOrderNum
Browse the API Results and assign the field SalesOrder



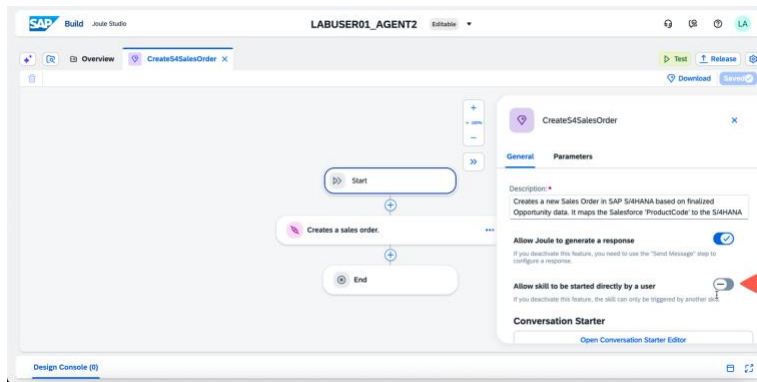
104.



105.
106.

Click Save

- 107. Next, we want to click on 'Trigger' and disable this skill from being called directly. This will ensure that Joule does have issues with intent matching.
- 108. Deselect the button below and click Save. The trigger button will change to a Start Button

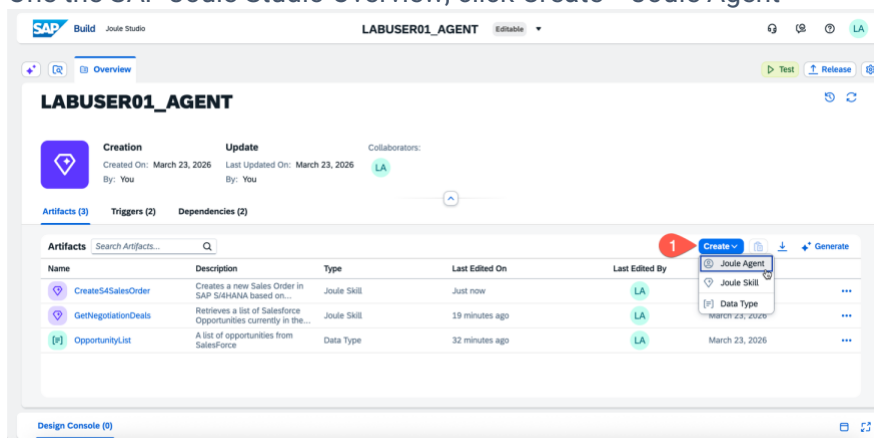


109.
110.

You're ready for testing!

Create a Joule Agent

1. One the SAP Joule Studio Overview, click Create > Joule Agent



- 2.
3. Enter the following details

- a. Name: SalesAssistant
- b. Description: Acts as a Sales Assistant to bridge Salesforce and SAP S/4HANA. It identifies Salesforce Opportunities currently in the 'Negotiation/Review' stage, retrieves their associated product line items (including Product Codes and Quantities), and automates the creation of corresponding Sales Orders in the SAP S/4HANA system. Use this agent when a user wants to sync finalized deal data from CRM to ERP or check for pending sales transitions.

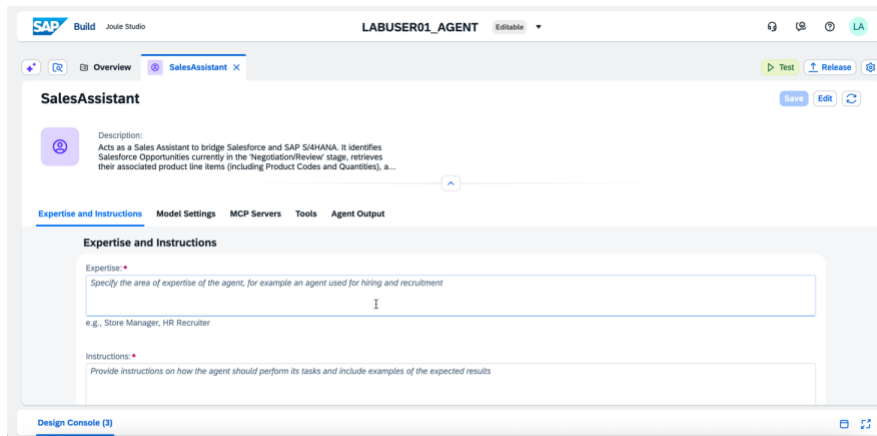
Edit information

Name: *

Identifier: *

Description: *

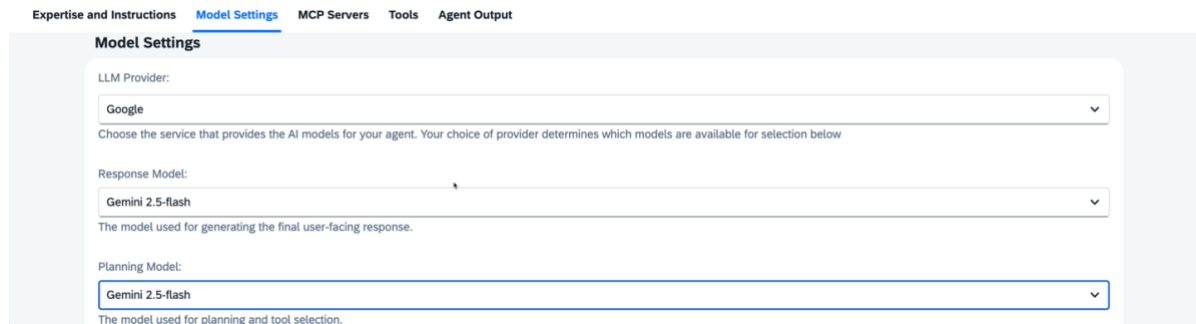
- 4.
5. On the Joule Agent screen, enter the following text notes



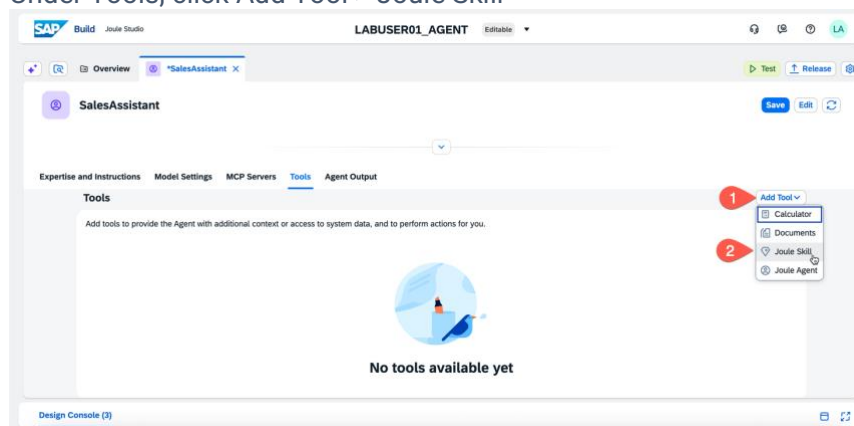
6.
7. Text Detail

Expertise	Expert in Salesforce CRM Opportunity Management, specifically the 'Negotiation/Review' sales stage and related Opportunity Line Items. Proficient in SAP S/4HANA Sales Order processing (API_SALES_ORDER). Skilled in data mapping between CRM Product Codes and ERP Material Numbers, ensuring quantity and unit price alignment during the lead-to-order transition.
Instructions	Trigger: When a user asks to check Salesforce opportunities, call the GetNegotiationDeals action which will get a list of Opportunities and associated line items by Opportunity Name which includes SoldToParty as SAP_Sold_To_Party__c in the response. Data Retrieval: Extract the Opportunity Name, Amount, Account Name, SoldToParty and the nested OpportunityLineItems (specifically ProductCode and Quantity). Validation: Before creating an order in S/4HANA, display the list of found products to the user and ask: "I found these opportunities with items in Salesforce. Should I proceed with creating the S/4HANA Sales Order?" Mapping from Salesforce output to SAP required inputs for CreateS4SalesOrder: 1) Name to CustomerRef 2) ProductCode to Material 3) Set RegQtyUnit to 'PC' 4) Quantity to RequestedQty 5) SAP_Sold_To_Party__c to SoldToParty Execution: Once confirmed, call the S/4HANA Sales Order creation API and confirm with the resulting sales order number (salesordernum) Confirmation: Provide the user with the new S/4HANA Sales Order Number.
Additional Context	<blank>

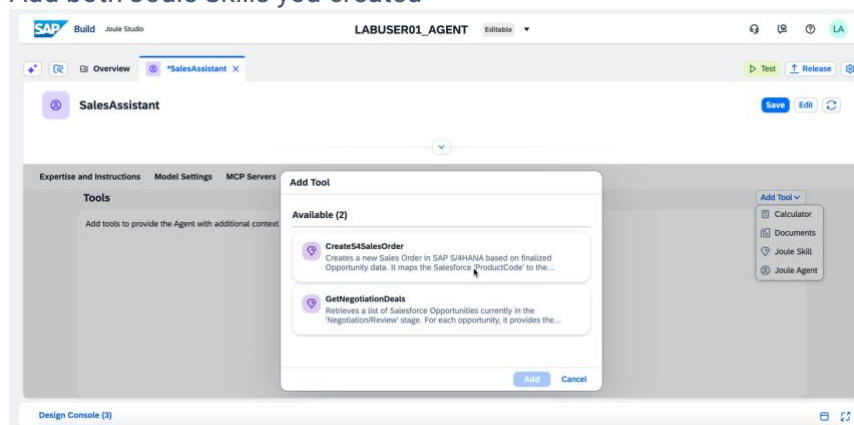
8. For Model Settings, select the following



- 9.
10. Leave all other Model Settings as Default
11. Skip MCP Servers
12. Under Tools, click Add Tool > Joule Skill



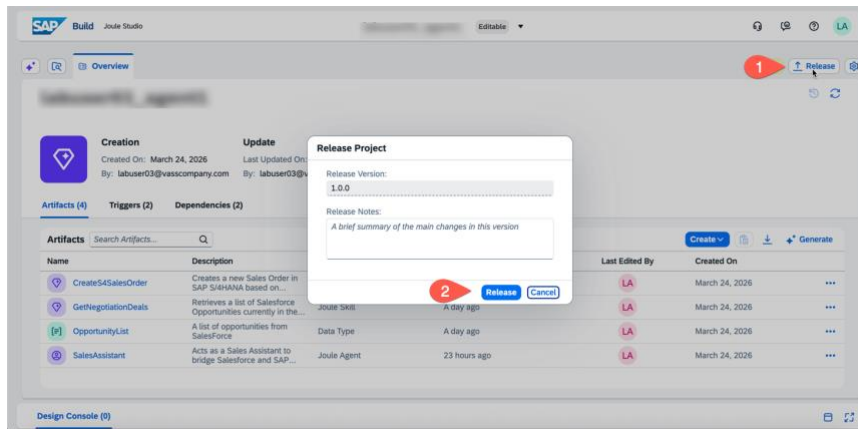
- 13.
14. Add both Joule Skills you created



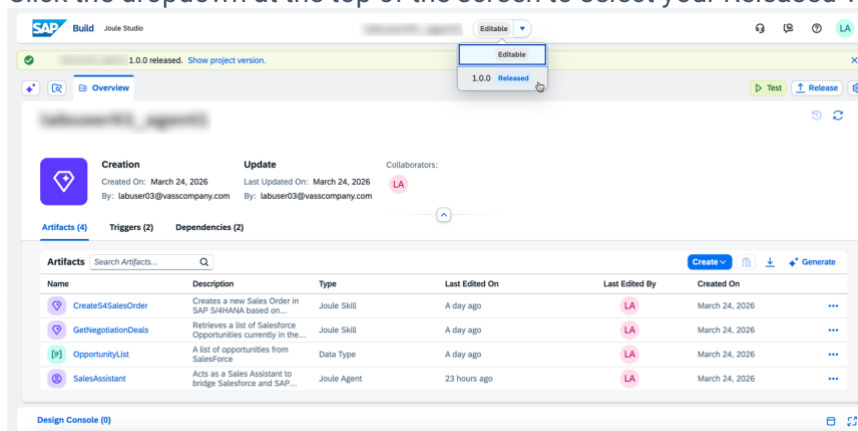
- 15.
16. Leave the defaults under Agent Output
17. Click Save
18. You're now ready to Test!

Test your Joule Agent

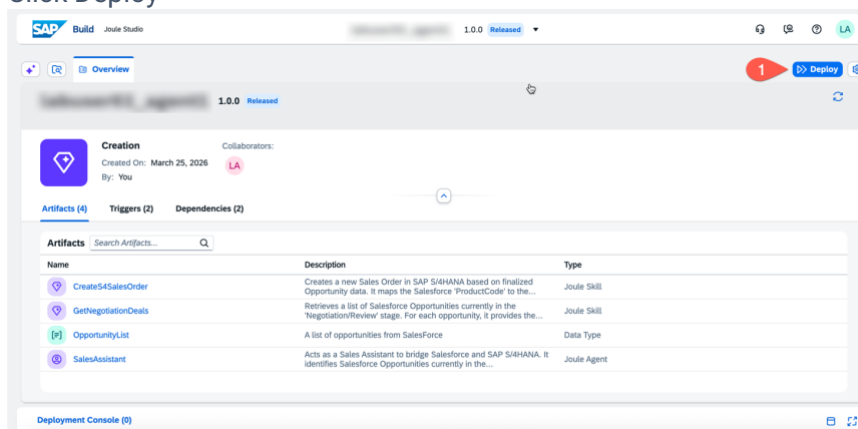
1. Under normal development conditions, a developer can click the "Test Button" in the upper right. In this lab session, multiple developers are developing the same Skills and Agent. When/If we clicked "Test" then every Skill would be pushed to the same environment. To separate our developments, we need to Release and Publish our Joule Skills/Agent to separate environments.
2. Click the Release Button in the upper right; then click Release



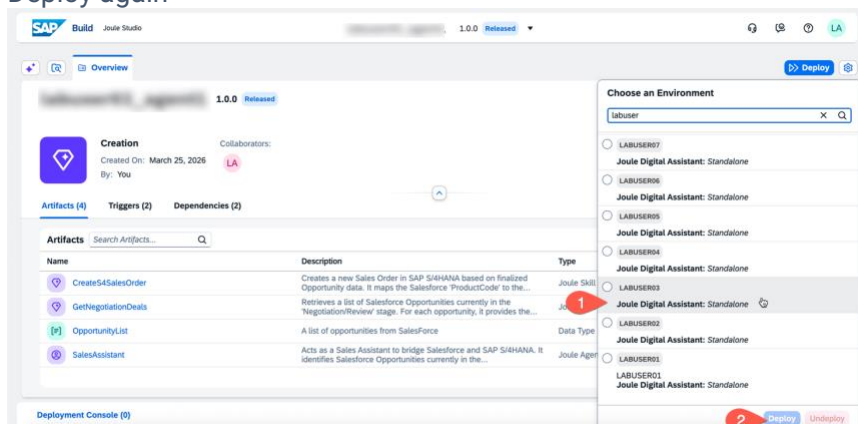
3. Click the dropdown at the top of the screen to select your Released version



5. Click Deploy

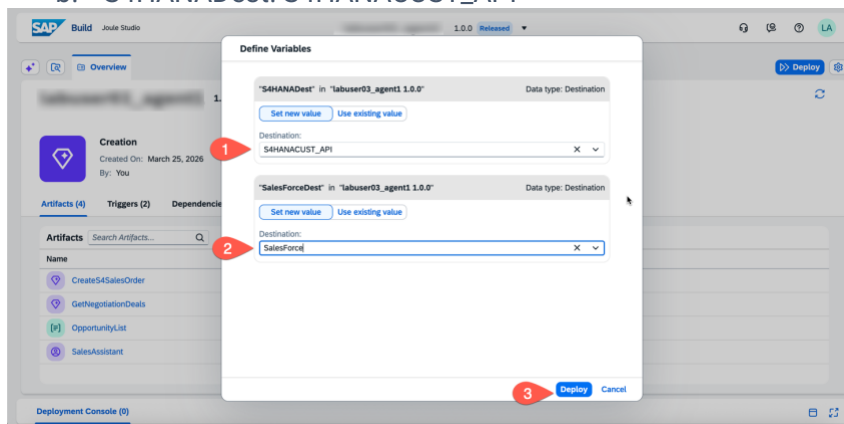


7. **IMPORTANT:** Select the environment that corresponds to your LABUSERXX ID and then click Deploy again

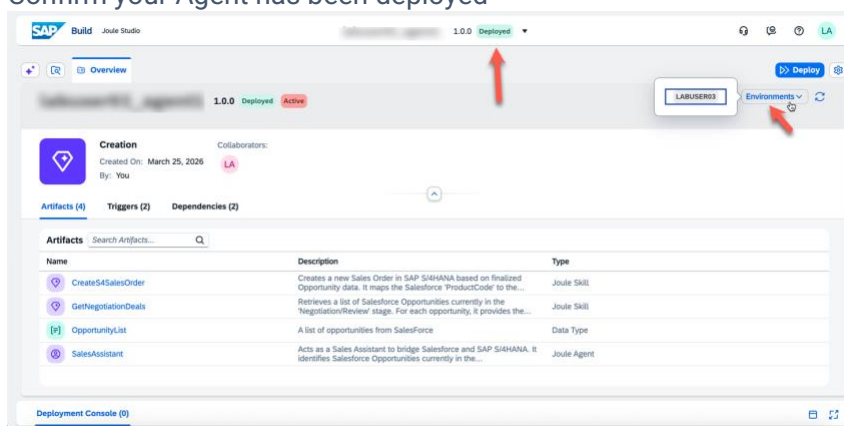


9. Enter the Destinations as shown and click Deploy

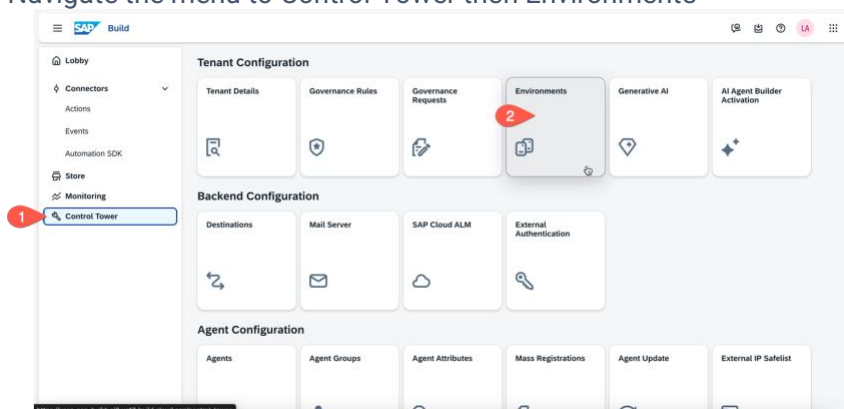
- a. SalesforceDest: Salesforce
- b. S4HANADest: S4HANACUST_API



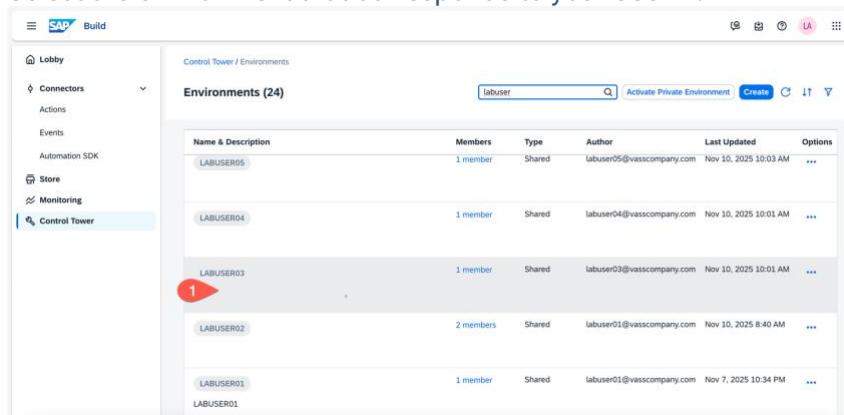
- 12.
13. Confirm your Agent has been deployed



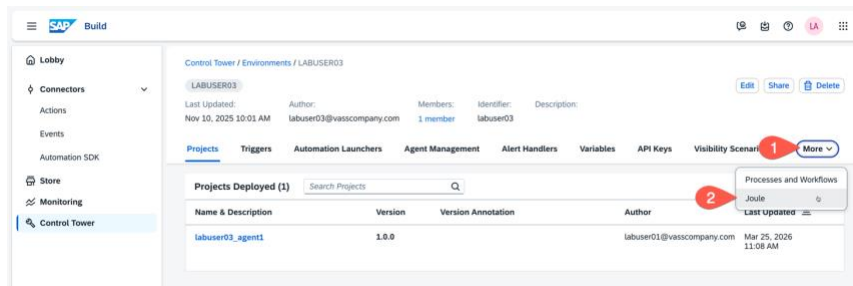
- 14.
15. Return to the SAP Build Lobby ([link](#)) on the prior browser tab
16. Navigate the menu to Control Tower then Environments



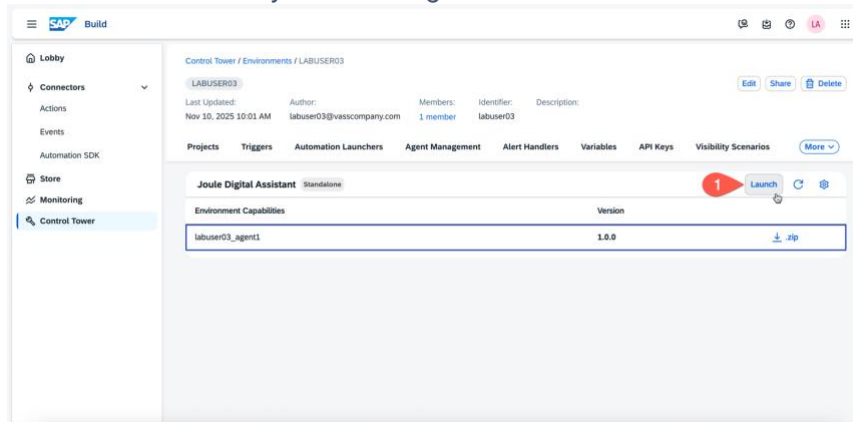
- 17.
18. Select the environment that corresponds to your UserID.



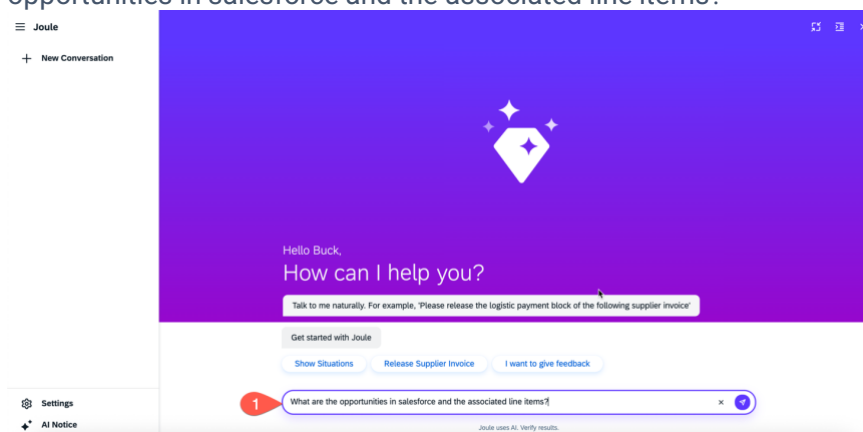
- 19.
20. Click the More button and then select Joule



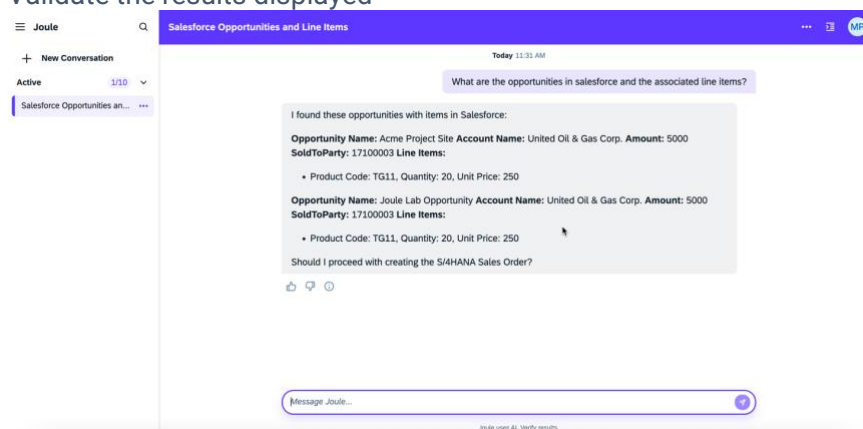
21. Click 'Launch' to test your Joule Agent



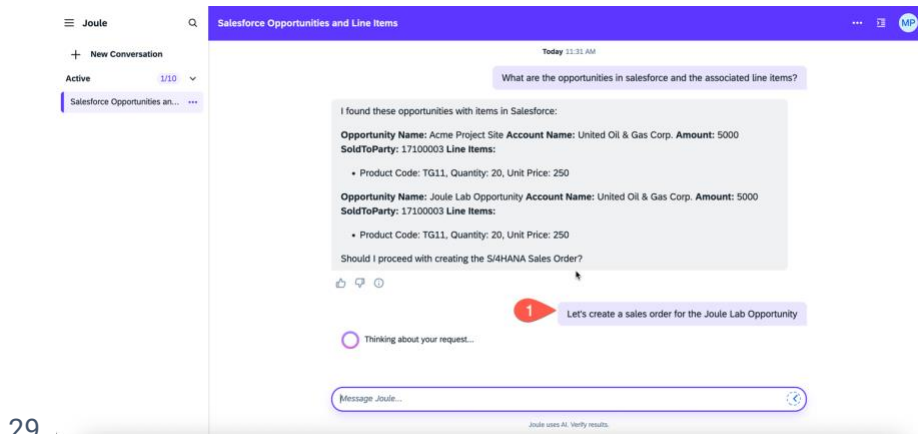
23. Enter a request that aligns with the instructions we gave above. For example, "What are the opportunities in salesforce and the associated line items?"



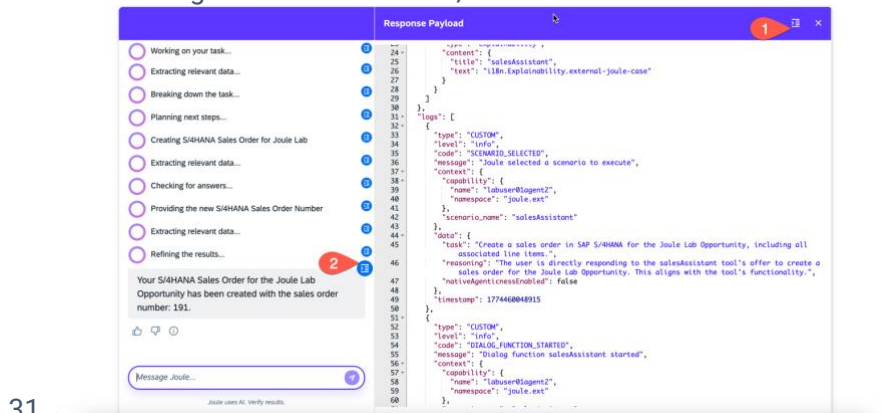
25. Validate the results displayed



27. Next, test the S/4HANA Sales Order entry with "Lets create a sales order for the XYZ Opportunity"; choose any opportunity you see.



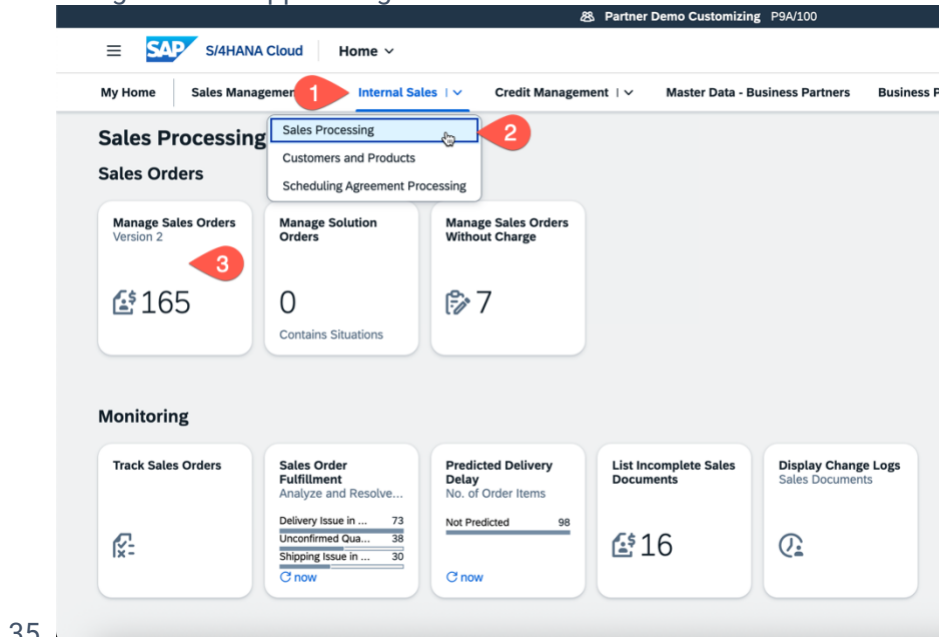
29. To view Logs for the transaction, click the buttons shown below.



31. And confirm the results in S/4HANA!

32. Logon to <https://my421457.s4hana.cloud.sap/>

33. Navigate to the app Manage Sales Orders



35.

36.

Sales Order	Sold-to Party	Customer Reference	Overall Status	Requested Delivery Date	Net Value	Document
178	Domestic Customer US 3 (17100003)	Labuser01 Opportunity	Open	05/27/2025	5,000.00 USD	03/24/2026
177	Domestic Customer US 3 (17100003)	PO 9876	Open	05/27/2025	5,000.00 USD	03/24/2026
176	Domestic Customer US 3 (17100003)	PO 6789	Open	05/27/2025	5,000.00 USD	03/24/2026
175	Domestic Customer US 3 (17100003)	PO 34567	Open	05/27/2025	5,000.00 USD	03/24/2026
174	Domestic Customer US 3 (17100003)	PO 12345	Open	05/27/2025	5,000.00 USD	03/24/2026
173	Domestic Customer US 3 (17100003)	Customer PO	Open	05/27/2025	5,000.00 USD	03/23/2026
172	Domestic Customer US 3 (17100003)	Customer PO	Open	05/27/2025	5,000.00 USD	03/23/2026
171	Domestic Customer US 3 (17100003)	Customer PO	Open	05/27/2025	0.00 USD	03/23/2026
170	Domestic Customer US 3 (17100003)	Customer PO	Open	05/27/2025	0.00 USD	03/23/2026
169	Domestic Customer US 3 (17100003)	Customer PO	Not Relevant	05/27/2025	0.00 USD	03/23/2026

37.

Standard Order 178

Sales Order: 178 Overall Status: Open Oncl Delivery Status: Not Delivered Approval Status: Not Relevant Net Value: 5,000.00 USD

Sold-to Party: Domestic Customer US 3 (17100003) Rejection Status: Nothing Rejected Credit Status: Not Performed

Ship-to Party: Domestic Customer US 3 (17100003)

General Information

Sales Area: US Sales Org., Distribution Channel, Division	Customer Reference: Labuser01 Opportunity	Document Date: 03/24/2026	Created By: Technical user for API_BILLINGDOCUMENT (CC000000000004)
Employee Responsible: --	Customer Reference Date: --	Billing Block: --	Created On: 03/24/2026
		Delivery Block: --	Last Changed By: Technical user for API_BILLINGDOCUMENT (CC000000000004)
			Changed On/At: 03/23/2026, 22:18:26

Thank you for completing our Hands-On Lab! Please email if you have any questions!